Orbit Equivalence

We will assume that (X, μ) is a nonatomic standard Borel probability space (i.e., a Borel probability space over a separable completely metrizable space X).

Basics and Definitions

Definition: A *countable Borel equivalence relation* $\mathcal{R} \subseteq X \times X$ is a Borel subset and equivalence relation satisfying the assumption that

$$[x]_{\mathcal{R}} \coloneqq \{y \in X \mid (x, y) \in \mathcal{R}\}$$

is countable.

The set $[\mathcal{R}]$ is known as the *full group* of the equivalence relation $\mathcal{R}$ and consists of all Borel isomorphisms $\phi\colon X \to X$ with its graph satisfying $\operatorname{graph}(\phi) \subseteq \mathcal{R}$.

The set of all *partial* Borel isomorphisms $\phi\colon \operatorname{dom}(\phi) \to \operatorname{ran}(\phi)$ with $\operatorname{graph}(\phi) \subseteq \mathcal{R}$ will be denoted $[[\mathcal{R}]]$.

The most basic example of a countable Borel equivalence relation is the space of orbits from the action of a countable discrete group $\Gamma \curvearrowright X$. We will write this as $\mathcal{R}(\Gamma \curvearrowright X)$. From now on, we will use "equivalence relation" to refer to countable Borel equivalence relations.

In fact, one can show that all such equivalence relations arise in this form.

Theorem (Feldman–Moore Theorem)**:** If $\mathcal{R}$ is a countable Borel equivalence relation on X, then there exists a countable group Γ and a Borel action $\Gamma \curvearrowright X$ such that $\mathcal{R} = \mathcal{R}(\Gamma \curvearrowright X)$. Moreover, both Γ and the action can be chosen such that $(x, y) \in \mathcal{R}$ if and only if there is $g \in \Gamma$ with order 2 such that $y = g \cdot x$.

We say the measure μ on X is $\mathcal{R}$-invariant if

$$\phi_* \mu = \mu$$

for all $\phi \in [\mathcal{R}]$, where $\phi_* \mu(U) = \mu\big(\phi^{-1}(U)\big)$ for all Borel $U \subseteq X$.

Proposition: The following are equivalent:

(a) μ is $\mathcal{R}$-invariant;

(b) μ is invariant under any Γ where Γ is such that $\mathcal{R} = \mathcal{R}(\Gamma \curvearrowright X)$;

(c) μ is Γ-invariant for some Γ with $\mathcal{R} = \mathcal{R}(\Gamma \curvearrowright X)$;

(d) for all $\phi \in [[\mathcal{R}]]$, we have $\mu(\operatorname{dom}(\phi)) = \mu(\operatorname{ran}(\phi))$.

Proof. We have (d) implies (a) since $[[\mathcal{R}]] \supseteq [\mathcal{R}]$, while (a) implies (b) since the action of g is an element of $[\mathcal{R}]$, and (b) implies (c) by definition. To see (c) implies (d), if $A = \operatorname{dom}(\phi)$ and $B = \operatorname{ran}(\phi)$, then there is a Borel partition $A = \bigcup_{n \geq 1} A_n$ and some $\{\gamma_n\}_{n \geq 1}$ such that for any $x \in A_n$, we have $\phi(x) = \gamma_n \cdot x$, meaning $B = \bigcup_{n \geq 1} \gamma_n \cdot A_n$. $\square$

Definition: A measure μ is called $\mathcal{R}$-ergodic if every $\mathcal{R}$-invariant Borel set is either null or co-null. The measure μ is ergodic with respect to the action of a group Γ if it is ergodic for $\mathcal{R}(\Gamma \curvearrowright X)$.

We discuss an example.

Let μ be the product measure on the space $X = 2^{\Gamma}$, where Γ is a countably infinite group, and $2 = \{0, 1\}$ assigns weight $1/2$ to each element. For each finite $S \subseteq \Gamma$, and map $s\colon S \to 2$, set

$$\mathcal{N}_s = \{f \mid f|_S = s\}.$$

Then, the set $\mathcal{N}_s$ forms a clopen basis for 2^{Γ}, and we may define $\mu(\mathcal{N}_s) = 2^{-|S|}$. Since

$$\gamma \cdot \mathcal{N}_s = \mathcal{N}_{\gamma \cdot s},$$

we have that μ is shift-invariant.

In fact, we claim that μ is mixing, in that

$$\lim_{n\to\infty} \mu(\gamma \cdot (A \cap B)) = \mu(A)\mu(B)$$

for any Borel $A, B \subseteq X$. Given any $A, B \subseteq X$ and $\varepsilon > 0$, we may find finite unions of clopen sets A' and B' such that $\mu(A \triangle A'), \mu(B \triangle B') < \varepsilon$. In particular, this means we only need to show the mixing condition when A and B are finite unions of basic clopen sets. If A is supported on S and B is supported on T, where S, T are finite, then for all but finitely many γ, we have $\gamma \cdot S \cap T = \emptyset$, meaning $\gamma \cdot A$ and B are independent, meaning

$$\begin{aligned}\mu(\gamma \cdot A \cap B) &= \mu(\gamma \cdot A)\mu(B)\\ &= \mu(A)\mu(B).\end{aligned}$$

Definition: We say $(X, \mathcal{R})$ and $(Y, \mathcal{S})$ are Borel isomorphic if there is a Borel bijection $\pi\colon X \to Y$ such that $(x, y) \in \mathcal{R}$ if and only if $(\pi(x), \pi(y)) \in \mathcal{S}$.

If μ and ν are invariant Borel probability measures on X and Y respectively, then we say they are almost everywhere Borel isomorphic if there are conull invariant Borel subsets $A \subseteq X$ and $B \subseteq Y$ and a Borel isomorphism $\pi\colon (A, \mathcal{R}|_A) \to (B, \mathcal{S}|_B)$ with $\pi_*\mu = \nu$.

Definition: We say two actions $\Gamma \curvearrowright X$ and $\Lambda \curvearrowright Y$ are *orbit equivalent* if the respective orbit equivalence relations $\mathcal{R}(\Gamma \curvearrowright X)$ and $\mathcal{R}(\Lambda \curvearrowright Y)$ are Borel isomorphic.

If the actions of Γ and Λ are both measure-preserving actions on finite Borel measure spaces, then they are orbit equivalent if the equivalence relations are almost everywhere Borel isomorphic with $\pi_*\mu = \nu$.

Von Neumann Algebras of Equivalence Relations

We start by considering a type of "mass transport principle" on an equivalence relation $\mathcal{R} \subseteq X \times X$, where (X, μ) is some probability space. From now on, we will disregard null sets (i.e., "every" means "almost every"). We will also assume that (almost) every orbit of $\mathcal{R}$ is infinite.

Definition: If $\mathcal{R} \subseteq X \times X$ is an equivalence relation, and $\mathcal{W} \subseteq \mathcal{R}$ is Borel, then we define

$$\nu(\mathcal{W}) = \int_X \mu(\{y \in X \mid (x, y) \in \mathcal{W}\})\, d\mu(x).$$

Note that since μ is invariant under $\mathcal{R}$, we have that

$$\int_X \mu(\{y \in X \mid (x, y) \in \mathcal{W}\})\, d\mu(x) = \int_X \mu(\{x \in X \mid (x, y) \in \mathcal{W}\})\, d\mu(y).$$

Note that if $f\colon X_1 \to X_2$ is a Borel isomorphism, then there is a unique $\pi_f\colon L_\infty(X_1) \to L_\infty(X_2)$ given by $\pi_f(\phi)(x) = \phi\big(f^{-1}(x)\big)$.

Now, consider the space $L_2(\mathcal{R}, \nu)$. If $\phi \in [[\mathcal{R}]]$, and $\xi \in L_2(\mathcal{R}, \nu)$, define

$$(u_\phi \xi)(x, y) = \chi_{\operatorname{ran}(\phi)}(x)\xi\big(\phi^{-1}(x), y\big).$$

That is,

$$(u_\phi \xi)(x, y) = \xi\big(\phi^{-1}(x), y\big)$$

if $x \in \operatorname{ran}(\phi)$ and 0 otherwise. The von Neumann algebra $L_\infty(X)$ acts on $L_2(\mathcal{R}, \nu)$ by multiplication in the x coordinate:

$$(F\xi)(x, y) = F(x)\xi(x, y).$$

If $\phi \in [[\mathcal{R}]]$ and $F \in L_\infty(X)$, then $F_\phi \in L_\infty(X)$ can be defined by

$$F_\phi(x) = \chi_{\operatorname{ran}(\phi)}(x)F\big(\phi^{-1}(x)\big).$$

That is, $F_\phi(x) = F\big(\phi^{-1}(x)\big)$ if $x \in \operatorname{ran}(\phi)$ and 0 otherwise. We call this the left-regular representation of $\mathcal{R}$.

Similarly, it is possible to define the right-regular representation of $\mathcal{R}$ by taking $(v_\phi \xi)(x, y) = \mathbb{1}_{\operatorname{ran}(\phi)}(y)\xi\big(x, \phi^{-1}(y)\big)$.

Proposition: If $\phi, \psi \in [[R]]$, the following hold:

(i) $u_\phi u_\psi = u_{\phi\psi}$;

(ii) $u_\phi^* = u_{\phi^{-1}}$;

(iii) $u_\phi^* u_\psi = \chi_{\mathrm{dom}(\phi)}$ and $u_\phi u_\phi^* = \chi_{\mathrm{ran}(\phi)}$;

(iv) $u_\phi F = F_\phi u_\phi$ for any $F \in L_\infty(X)$.

Definition: Define the von Neumann algebra $L(\mathcal{R}) \subseteq B(L_2(\mathcal{R}, \nu))$ to be the von Neumann algebra generated by the set $\{u_\phi \mid \phi \in [[\mathcal{R}]]\}$.

Likewise, define $R(\mathcal{R}) \subseteq B(L_2(\mathcal{R}, \nu))$ to be the von Neumann algebra generated by $\{v_\phi \mid \phi \in [[\mathcal{R}]]\}$.

Before we may start understanding the von Neumann algebra of such equivalence relations, we discuss the standard representation of a finite von Neumann algebra (M, τ).

Proposition: Let M be a finite von Neumann algebra with trace τ. Then, the map $x \mapsto x^*$ extends to an anti-linear isometry $J\colon L_2(M, \tau) \to L_2(M, \tau)$ with $JMJ = M' \cap B(L_2(M, \tau))$. Here, $L_2(M, \tau)$ refers to the GNS representation of M with respect to the faithful state τ.

Proof. For any $x \in M$, we have

$$\begin{aligned}\|x^*\|_\tau^2 &= \tau(xx^*)\\ &= \tau(x^*x)\\ &= \|x\|_\tau^2.\end{aligned}$$

Thus, this extends to an anti-linear isometry $J\colon L_2(M, \tau) \to L_2(M, \tau)$. We have $\langle J\xi, J\eta\rangle = \langle \xi, \eta\rangle$ for all $\xi, \eta \in L_2(M, \tau)$.

For any $\xi \in L_2(M, \tau)$, define the unbounded operators $L_\xi^0\colon M1_\tau \to L_2(M, \tau)$ and $R_\xi^0\colon M1_\tau \to L_2(M, \tau)$ by

$$\begin{aligned}L_\xi^0(x1_\tau) &= (Jx^*J)\xi\\ R_\xi^0(x1_\tau) &= x\xi.\end{aligned}$$

Here, 1_τ refers to the cyclic vector for the representation τ.

If $(x_i)_i \in M$ and $\eta \in L_2(M, \tau)$ are such that $\|x_i\|_\tau \to 0$ and $L_\xi^0(x_i 1_\tau) \to \eta$, then for all $z \in M$, we have

$$\begin{aligned}|\langle \eta, z\rangle| &= \lim_i |\langle (Jx_i^*J)\xi, z\rangle|\\ &= \lim_i |\langle \xi, zx_i^*\rangle|\\ &\leq \lim_i \|\xi\|_\tau \|z\|_{\mathrm{op}} \|x_i\|_\tau\\ &= 0,\end{aligned}$$

meaning that L_ξ^0 is closable. Completing the graph gives rise to a closed operator L_ξ, and similarly R_ξ^0 is closable with corresponding operator R_ξ. We have that $L_{x1_\tau} = x$ for all $x \in M$.

Now, if $x, y \in M$ and $\xi \in L_2(M, \tau)$, then

$$\begin{aligned}\langle R_{J\xi}(x1_\tau), y1_\tau\rangle &= \langle xJ\xi, y1_\tau\rangle\\ &= \langle y^*x1_\tau, \xi\rangle\\ &= \langle x1_\tau, R_\xi(y1_\tau)\rangle,\end{aligned}$$

meaning that $R_{J\xi} = R_\xi^*$. Additionally,

$$\begin{aligned}(JL_\xi J)(x1_\tau) &= xJ\xi\\ &= R_{J\xi}(x1_\tau)\end{aligned}$$

$$= R_\xi^*(x1_\tau),$$

so that $JL_\xi J = R_\xi^*$.

If we set $\widetilde{M} = \{L_\xi \mid L_\xi \in B(L_2(M,\tau))\}$ and $\widetilde{N} = \{R_\xi \mid R_\xi \in B(L_2(M,\tau))\}$, then we have $J\widetilde{M}J = \widetilde{N}$.

Therefore, it suffices to show that $M = \widetilde{M}$ and $M' = \widetilde{N}$. For this, we observe that $M \subseteq \widetilde{M}$ and $\widetilde{N} \subseteq M'$. For any $x \in M'$, let $\xi = x1_\tau$. For any $y \in M$, we then have

$$\begin{aligned} R_\xi(y1_\tau) &= y\xi \\ &= yx1_\tau \\ &= xy1_\tau, \end{aligned}$$

meaning that $R_\xi = x$, so $M' \subseteq \widetilde{N}$. For any $L_\xi \in \widetilde{M}$, we then have $L_\xi \in \widetilde{N}' = M'' = M$, so $\widetilde{M} \subseteq M$. □

We will specialize this for the case of the von Neumann algebra $L(\mathcal{R})$. Observe that $L(\mathcal{R}) \subseteq R(\mathcal{R})'$, so we may define the canonical anti-unitary $J\colon L_2(\mathcal{R},\nu) \to L_2(\mathcal{R},\nu)$ by $(J\xi)(x,y) = \overline{\xi(y,x)}$.

Proposition: Let $\xi_0 = \chi_\Delta \in L_2(\mathcal{R},\nu)$, where $\Delta \subseteq \mathcal{R}$ is the diagonal.

(i) We have that ξ_0 is a cyclic and separating vector for $L(\mathcal{R})$.

(ii) The vector state $\tau = \langle \cdot \xi_0, \xi_0 \rangle$ is a faithful normal trace on $L(\mathcal{R})$. In particular, $L(\mathcal{R})$ is a finite von Neumann algebra.

(iii) For any $\phi \in [[\mathcal{R}]]$, we have $Ju_\phi J = v_\phi$, and in particular, $L(\mathcal{R}) = R(\mathcal{R})'$.

Proof.

(i) Let $\xi_0 = \chi_\Delta$. For any $\phi \in [[\mathcal{R}]]$, we have

$$\chi_{\text{graph}(\phi)} = u_{\phi^{-1}}\xi_0.$$

Since $\mathcal{R}$ can be written as a countable disjoint union of the graphs of Borel isomorphisms, $\mathcal{R} = \bigsqcup_n \text{graph}(g_n)$, so we may write any $W \subseteq \mathcal{R}$ as $W = \bigsqcup_n W_n$ with $W_n = W \cap \text{graph}(g_n)$. Since W_n is the graph of a partial Borel isomorphism, we have that $\chi_W \in \overline{L(\mathcal{R})\xi_0}$, so ξ_0 is cyclic for $L(\mathcal{R})$.

□

Amenability and Hyperfiniteness

Our goal now is to show that non-atomic probability measure preserving ergodic actions of amenable groups are orbit equivalent. This is a conclusion of the combination of two theorems; a theorem of Dye says that any two non-atomic probability measure-preserving ergodic actions of $\mathbb{Z}$ are orbit equivalent, and a theorem of Ornstein and Weiss says that any probability measure-preserving action of an amenable group is orbit equivalent to one of $\mathbb{Z}$. This will eventually yield the celebrated Connes–Feldman–Weiss theorem, which states the following.

Theorem: Let $\mathcal{R}$ be a countable Borel equivalence relation on X, and let μ be an $\mathcal{R}$-quasi-invariant probability measure. If $\mathcal{R}$ is μ-amenable, then $\mathcal{R}$ is almost everywhere hyperfinite.

We will assume command of understanding of amenability in the case of groups, which is discussed in depth in this document.

Hyperfiniteness

We start by studying equivalence relations induced by $\mathbb{Z}$-actions.

Definition: If $\mathcal{R}$ is a Borel equivalence relation on X, we say $\mathcal{R}$ is *smooth* if it admits a family $\{B_n\}_{n\in\mathbb{N}}$ of Borel subsets of X such that for all $x, y \in X$, $(x,y) \in \mathcal{R}$ if and only if, for all $n \in \mathbb{N}$, $x \in B_n$ if and only if $y \in B_n$.

Via the maps $x \mapsto \{\chi_{B_n}(x)\}_{n\in\mathbb{N}}$, we see that this definition says $\mathcal{R}$ is smooth if and only if there is a Borel function $\phi\colon X \to Y$ with Y a standard Borel space such that for all $x, y \in X$, we have $(x, y) \in \mathcal{R}$ if and only if $\phi(x) = \phi(y)$.

Definition: If $\mathcal{R}$ is a Borel equivalence relation, we define $X/\mathcal{R}$ to be the set of equivalence classes of $\mathcal{R}$ equipped with the quotient Borel structure, wherein $A \subseteq X/\mathcal{R}$ is Borel if and only if $\bigcup A$ is Borel in X.

Proposition: A countable Borel equivalence relation $\mathcal{R}$ is smooth if and only if $X/\mathcal{R}$ is standard Borel.

Proof. Let $\phi\colon X \to Y$ be a surjection that gives the smoothness of $\mathcal{R}$, and let φ be the induced bijection from $X/\mathcal{R} \to Y$. It suffices to verify that φ induces an isomorphism o the quotient Borel structure $X/\mathcal{R}$ with the standard Borel structure of Y. Since ϕ is countable-to-one, we have that ϕ maps Borel sets to Borel sets, meaning that an $\mathcal{R}$-invariant set B is Borel if and only if $\varphi(B) = \phi(B)$ is Borel.

In the reverse direction, if the quotient Borel structure on $X/\mathcal{R}$ is standard Borel, then there is a countable family of Borel sets $\{B_n\}_{n\in\mathbb{N}}$ in the quotient Borel structure that separates points. Yet, any such family induces a Borel separating family for $\mathcal{R}$. □

Definition: A *complete section* for $\mathcal{R}$ is a set $B \subseteq X$ that intersects every equivalence class in $\mathcal{R}$.

A *transversal* for $\mathcal{R}$ is a set $B \subseteq X$ that intersects every equivalence class in $\mathcal{R}$ exactly once.

A *selector* for $\mathcal{R}$ is a function $f\colon X \to X$ with $\operatorname{graph}(f) \subseteq \mathcal{R}$ and $f(X) \subseteq X$ is a transversal.

A Borel equivalence relation admits a Borel transversal if and only if it admits a Borel selector; if it admits a Borel transversal, then it is smooth (but this is not true in reverse).

Proposition: If $\mathcal{R}$ is a *countable* smooth equivalence relation, then $\mathcal{R}$ admits a Borel selector. Moreover, if all the classes of $\mathcal{R}$ are of size $n \in \mathbb{N} \cup \{\infty\}$, then there is a sequence $\{f_i\}_{i<n}$ of Borel selectors for $\mathcal{R}$ whose images partition X.

Proof. Let $\phi\colon X \to Y$ be a surjection that implements the smoothness of $\mathcal{R}$, and define a relation $\mathcal{S} \subseteq Y \times X$ by saying $(y, x) \in \mathcal{S}$ if and only if $\phi(x) = y$.

Since $\mathcal{S}$ is Borel and has countable sections, it follows from the Lusin–Novikov Theorem that there is a Borel uniformization $f\colon Y \to X$ of $\mathcal{S}$. It follows that $f \circ \phi$ is our desired Borel selector.

Now, if all the classes of $\mathcal{R}$ are of size $n \leq \aleph_0$, there are then sequences $\{f_i\}_{i<n}$ of Borel uniformizations of $\mathcal{S}$ with disjoint ranges whose union is X, so $\{f_i \circ \phi\}_{i<n}$ is a sequence of Borel selectors whose images partition X. □

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